E-Commerce Order Management Database System

1. Introduction

The E-Commerce Order Management Database System is designed to manage all activities involved in an online shopping platform. The system stores customer information, product details, inventory, supplier information, orders, payments, shipments, and customer reviews in a centralized database. This improves data accuracy, reduces duplication, and increases operational efficiency.

2. Business Scenario Analysis

The company is expanding rapidly and needs a centralized database system to manage its daily operations.

The proposed system should:

Register customers

Store product information

Manage product categories

Track stock quantity

Process customer orders

Record payments

Track shipments

Store customer reviews

Generate business reports

The database will help improve customer service, inventory management, and business decision-making.

3. Purpose of Core Entities

Customer

Stores customer information such as name, email, mobile number, address, and registration date.

Category

Stores product categories to organize products.

Product

Stores product details including price, stock quantity, and category.

Supplier

Stores supplier information for supplying products.

Order

Stores customer order information including order date, amount, and status.

Order Details

Stores products included in each order along with quantity and price.

Payment

Stores payment details such as payment method, payment date, and payment status.

Shipment

Stores delivery information and shipment status.

Review

Stores customer ratings and comments about purchased products.

4. Major Business Processes

- Customer Registration
- Product Management
- Category Management
- Inventory Management
- Supplier Management
- Order Placement
- Payment Processing
- Shipment Tracking
- Customer Review Management
- Report Generation

5. Non-Functional Requirements

- Easy to use
- Fast response time
- Secure customer information

- Reliable system
- High availability
- Data consistency
- Scalable database
- Regular backup
- User authentication
- Maintainable design

6. Project Scope

- The project includes:
- Customer Management
- Product Management
- Category Management
- Supplier Management
- Inventory Tracking
- Order Processing
- Payment Management
- Shipment Tracking
- Customer Reviews
- Business Reporting

7. Assumptions

Every customer has a unique Customer ID.

Every product belongs to one category.

Orders contain one or more products.

Payments are linked to orders.

Shipments are created after payment.

Customers review purchased products only.

8. Constraints

Only the given entities are used.

Internet connection is required.

Data must be stored in MySQL.

Unique IDs must be maintained.

No duplicate customer records.

9. Conclusion

The requirement analysis identifies all business needs, stakeholders, system functions, and project scope required for designing the E-Commerce Order Management Database System.